

Math 163, Spring 2026
SCRIPT 14: Integrals and Derivatives

Theorem 14.1. Suppose that f is integrable on $[a, b]$. Define $F: [a, b] \rightarrow \mathbb{R}$ by

$$F(x) = \int_a^x f.$$

If f is continuous at $p \in (a, b)$, then F is differentiable at p and

$$F'(p) = f(p).$$

If f is continuous at a , $F'_+(a)$ exists and equals $f(a)$. Similarly, if f is continuous at b , $F'_-(b)$ exists and equals $f(b)$.

Proof of Theorem 14.1. We want to show that

$$F'(p) = \lim_{x \rightarrow p} \frac{F(x) - F(p)}{x - p} = f(p).$$

For this, we can consider two cases.

Case 1: $x > p$. Let $x > p$. Then, we have that $\frac{F(x) - F(p)}{x - p}$ can be written as

$$\lim_{x \rightarrow p} \frac{F(x) - F(p)}{x - p} = \lim_{x \rightarrow p} \frac{\int_a^x f - \int_a^p f}{x - p} = \lim_{x \rightarrow p} \frac{\int_p^x f}{x - p}.$$

Since f is continuous at p , for any $\epsilon > 0$ there exists $\delta > 0$ such that if $|t - p| < \delta$ then $|f(t) - f(p)| < \epsilon$. In particular, for $x \in (p, p + \delta)$, we have $f(p) - \epsilon \leq f(t) \leq f(p) + \epsilon$ for all $t \in [p, x]$. Therefore, as f is integrable on $[a, b]$ and $m \leq f(x) \leq M$ for all $x \in [a, b]$, then by **Theorem 13.28** get that

$$(f(p) - \epsilon)(x - p) \leq \int_p^x f \leq (f(p) + \epsilon)(x - p).$$

Then, we can divide by $(x - p) > 0$ to get

$$f(p) - \epsilon \leq \frac{\int_p^x f}{x - p} \leq f(p) + \epsilon.$$

Hence we have that $|\lim_{x \rightarrow p} \frac{\int_p^x f}{x - p} - f(p)| < \epsilon$, which means that $\lim_{x \rightarrow p^+} \frac{F(x) - F(p)}{x - p} = f(p)$.

Case 2: $x < p$. Let $x < p$. We have

$$\lim_{x \rightarrow p} \frac{F(x) - F(p)}{x - p} = \lim_{x \rightarrow p} \frac{\int_a^x f - \int_a^p f}{x - p} = \lim_{x \rightarrow p} \frac{-\int_x^p f}{x - p} = \lim_{x \rightarrow p} \frac{\int_p^x f}{x - p}.$$

This is the same expression as in Case 1, and the same squeeze argument applies as $x \rightarrow p^-$, giving $\lim_{x \rightarrow p^-} \frac{F(x)-F(p)}{x-p} = f(p)$.

As $\lim_{x \rightarrow p^-} \frac{F(x)-F(p)}{x-p} = F'(p) = f(p)$ and $\lim_{x \rightarrow p^+} \frac{F(x)-F(p)}{x-p} = F'(p) = f(p)$ this suffices to show that $F'(p) = f(p)$. $\square$

Remark 14.2. Thus we have that if f is continuous on $[a, b]$, F is differentiable on $[a, b]$ and $F'(p) = f(p)$, for all $p \in [a, b]$, (where at the endpoints we understand that the derivative should be interpreted as the one-sided derivative).

Corollary 14.3. Let f be continuous on $[a, b]$. Suppose that there is a function G that is continuous on $[a, b]$ and differentiable on (a, b) and such that $f = G'$ on (a, b) . Then

$$\int_a^b f = G(b) - G(a).$$

Proof of Corollary 14.3. Let f be continuous on $[a, b]$. Suppose that there is a function G that is continuous on $[a, b]$ and differentiable on (a, b) and such that $f = G'$ on (a, b) . We want to show that $\int_a^b f = G(b) - G(a)$.

Let there be a function $F : [a, b] \rightarrow \mathbb{R}$ be defined as $F(x) = \int_a^x f$. As f is continuous on $[a, b]$ we have that, by **Theorem 14.1** F is differentiable on (a, b) . Then, we have that $F'(x) = f(x)$ for all $x \in [a, b]$.

Now, as $F'(x) = f(x)$ for all $x \in [a, b]$, we have that $F' = G'$. Then, by **Theorem 12.18**, we have that $F(x) = G(x) + c$ where $x \in \mathbb{R}$. Then, we can write

$$F(b) - F(a) = (G(b) + c) - (G(a) + c)$$

By definition of $F(x)$, we have that $F(x) = \int_a^x f$ and therefore we can rewrite this as

$$\int_a^b f - \int_a^a f = (G(b) + c) - (G(a) + c) = G(b) - G(a)$$

Thus, we get that

$$\int_a^b f = G(b) - G(a)$$

This suffices to complete the proof. $\square$

We now prove a more general version of this corollary, relaxing the condition on f from continuity to integrability. The following lemma will be useful.

Lemma 14.4. *Suppose that $f: [a, b] \rightarrow \mathbb{R}$ is integrable and that Ω is a number satisfying*

$$L(f, P) \leq \Omega \leq U(f, P) \quad \text{for every partition } P \text{ of } [a, b].$$

Then

$$\int_a^b f = \Omega.$$

Proof of Lemma 14.4. Suppose that $f: [a, b] \rightarrow \mathbb{R}$ is integrable and that Ω is a number satisfying $L(f, P) \leq \Omega \leq U(f, P)$ for every partition P of $[a, b]$. We want to show that $\int_a^b f = \Omega$.

Suppose for sake of contradiction that $\int_a^b f \neq \Omega$. Then it must be that either $\int_a^b f < \Omega$ or $\int_a^b f > \Omega$.

Firstly, consider the case that $\int_a^b f < \Omega$, then $U(f) = \int_a^b f < \Omega$. Since $\Omega > U(f)$, Ω is not a lower bound of the set of upper sums over all partitions P of $[a, b]$, so by definition of the infimum, there exists a partition P_0 of $[a, b]$ such that $U(f, P_0) < \Omega$. This contradicts the assumption that $L(f, P) \leq \Omega \leq U(f, P)$ for all partitions P of $[a, b]$.

Secondly, consider the case that $\int_a^b f > \Omega$, then $L(f) = \int_a^b f > \Omega$. Since $\Omega < L(f)$, Ω is not an upper bound of the set of lower sums over all partitions P of $[a, b]$, so by definition of the supremum, there exists a partition P_0 of $[a, b]$ such that $L(f, P_0) > \Omega$. This contradicts the assumption that $L(f, P) \leq \Omega \leq U(f, P)$ for all partitions P of $[a, b]$.

Hence $\int_a^b f = \Omega$. This completes the proof. □

Theorem 14.5. *Let f be integrable on $[a, b]$. Suppose that there is a function G that is continuous on $[a, b]$ and differentiable on (a, b) and such that $f = G'$ on (a, b) . Then*

$$\int_a^b f = G(b) - G(a).$$

Proof of Theorem 14.5. Let f be integrable on $[a, b]$. Then, suppose there is a function G that is continuous on $[a, b]$ and differentiable on (a, b) such that $f = G'$ on (a, b) . We want to show that $\int_a^b f = G(b) - G(a)$.

Now, let $\Omega = G(b) - G(a)$. By **Lemma 14.4**, it suffices to show that $L(f, P) \leq \Omega \leq U(f, P)$ for every partition P of $[a, b]$ in order to prove that $\int_a^b f = G(b) - G(a)$.

Let $P = \{t_0, t_1, \dots, t_n\}$ be any partition of $[a, b]$. We note that $G(b) - G(a) = \sum_{i=1}^n G(t_i) - G(t_{i-1})$ by telescoping. Then, by **Theorem 12.15**, for each $i \in [n]$ there exists $\lambda_i \in (t_{i-1}, t_i)$ such that $G(t_i) - G(t_{i-1}) = G'(\lambda_i)(t_i - t_{i-1}) = f(\lambda_i)(t_i - t_{i-1})$. Hence, $\Omega = G(b) - G(a) = \sum_{i=1}^n f(\lambda_i)(t_i - t_{i-1})$. Since $m_i(f) \leq f(\lambda_i) \leq M_i(f)$ for each $i \in [n]$, we have that $m_i(f)(t_i - t_{i-1}) \leq f(\lambda_i)(t_i - t_{i-1}) \leq M_i(f)(t_i - t_{i-1})$. When considering $L(f, P) = \sum_{i=1}^n m_i(f)(t_i - t_{i-1})$ and $U(f, P) = \sum_{i=1}^n M_i(f)(t_i - t_{i-1})$, we have that $L(f, P) \leq \Omega \leq U(f, P)$. As P was arbitrary, by **Lemma 14.4** we conclude $\int_a^b f = G(b) - G(a)$.

This completes the proof. $\square$

Corollary 14.6. *Let f, g be functions defined on some open interval containing $[a, b]$ such that f' and g' exist and are continuous on $[a, b]$. Then*

$$\int_a^b f g' = [f(b)g(b) - f(a)g(a)] - \int_a^b f' g.$$

Proof of Corollary 14.6. Consider $\int_a^b f g' + \int_a^b f' g$. We have that (by **Theorem 13.25**) $\int_a^b f g' + \int_a^b f' g = \int_a^b (f g)'$. Then, by **Theorem 14.3**, we have that there exists a function G that is continuous on $[a, b]$ and differentiable on (a, b) such that $(f g)' = G'$ on (a, b) . Then, we have that

$$\int_a^b (f g)' = G(b) - G(a) = [f(b)g(b) - f(a)g(a)]$$

Then, we can expand this to get

$$\int_a^b f' g + \int_a^b f g' = [f(b)g(b) - f(a)g(a)]$$

Finally, we can rearrange this to get

$$\int_a^b f g' = [f(b)g(b) - f(a)g(a)] - \int_a^b f' g$$

Thus, this suffices to complete the proof. $\square$

Corollary 14.7. *Let g be a function defined on some open interval containing $[a, b]$ such that g' is continuous on $[a, b]$. Suppose that $g([a, b]) \subset [c, d]$ and $f: [c, d] \rightarrow \mathbb{R}$ is continuous. Define $F: [c, d] \rightarrow \mathbb{R}$ by $F(x) = \int_c^x f$. Then*

$$\int_a^b f(g(x)) \cdot g'(x) dx = F(g(b)) - F(g(a)).$$

Proof of Corollary 14.7. Define $F: [c, d] \rightarrow \mathbb{R}$ by $F(x) = \int_c^x f$. Since f is continuous on $[c, d]$, by Theorem 14.1:

$$F'(x) = f(x) \quad \text{for all } x \in [c, d].$$

Consider the composition $F \circ g$. By the chain rule:

$$(F \circ g)'(x) = F'(g(x)) \cdot g'(x) = f(g(x)) \cdot g'(x).$$

Since g' is continuous on $[a, b]$ and f is continuous on $[c, d]$, the composition $f \circ g$ is continuous, and hence $f(g(x)) \cdot g'(x)$ is continuous and integrable on $[a, b]$. Furthermore, $F \circ g$ is continuous on $[a, b]$ and differentiable on (a, b) . By Corollary 14.3:

$$\int_a^b f(g(x)) \cdot g'(x) \, dx = \int_a^b (F \circ g)'(x) \, dx = (F \circ g)(b) - (F \circ g)(a) = F(g(b)) - F(g(a)).$$

□

Additional Exercises

1. What is wrong with the following argument? If $F(x) = \frac{1}{1-x}$ then $F'(x) = \frac{1}{(1-x)^2}$ and hence

$$\int_0^2 \frac{1}{(1-x)^2} dx = F(2) - F(0) = -2.$$

2. Assume f is integrable on $[a, b]$ and let $F : [a, b] \rightarrow \mathbb{R}$ be such that

$$F(x) = \int_a^x f(s) ds.$$

Give an example of an f for which F is *not* differentiable at p for some given $p \in (a, b)$.

3. Using integration by parts, compute Euler's integral

$$B(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx$$

for $m, n \in \mathbb{N}$.

Additional Exercise 3. Consider the expression $\int_0^1 x^{m-1}(1-x)^{n-1} dx$. We want to compute the integral (Euler's integral) using integration by parts.

Let $\int_0^1 x^{m-1}(1-x)^{n-1} dx$ for $m, n \in \mathbb{N}$. Then, through integration by parts, we can let $u = x^{m-1}$ and $dv = (1-x)^{n-1}$. Then, by **12.10**, we have that $du = (m-1)x^{m-2}$ and $v = -\frac{(1-x)^n}{n}$. We have that the formula for integration by parts is $\int u dv = uv - \int v du$. Therefore, by substituting these variations into the equation, we get

$$\int_0^1 x^{m-1}(1-x)^{n-1} dx = \left[-\frac{x^{m-1}(1-x)^n}{n} \right]_0^1 + \frac{m-1}{n} \int_0^1 (1-x)^n x^{m-2} dx.$$

We note that $\left[-\frac{x^{m-1}(1-x)^n}{n} \right]_0^1 = 0$, since at $x = 1$ the factor $(1-x)^n$ vanishes, and at $x = 0$ the factor x^{m-1} vanishes for $m \geq 2$. Furthermore, we note that $\frac{m-1}{n} \int_0^1 (1-x)^n x^{m-2} dx = \frac{m-1}{n} B(m-1, n+1)$, since by definition $B(m-1, n+1) = \int_0^1 x^{m-2}(1-x)^n dx$. Thus, we can write the integral as $B(m, n) = \frac{m-1}{n} \cdot B(m-1, n+1)$.

Then, consider $B(m, n)$. Applying the recurrence repeatedly, each time reducing the first argument by 1 and increasing the second by 1, after $m-1$ applications we obtain

$$B(m, n) = \frac{m-1}{n} \cdot \frac{m-2}{n+1} \cdot \frac{m-3}{n+2} \cdots \frac{1}{n+m-2} \cdot B(1, n+m-1) = \frac{(m-1)!}{n(n+1) \cdots (n+m-2)} \cdot B(1, n+m-1).$$

It remains to evaluate $B(1, n+m-1) = \int_0^1 (1-x)^{n+m-2} dx$. We have that

$$B(1, n+m-1) = \left[-\frac{(1-x)^{n+m-1}}{n+m-1} \right]_0^1 = 0 - \left(-\frac{1}{n+m-1} \right) = \frac{1}{n+m-1}.$$

Substituting back, we get

$$B(m, n) = \frac{(m-1)!}{n(n+1) \cdots (n+m-1)} = \frac{(m-1)!(n-1)!}{(m+n-1)!}.$$

This completes the proof. $\square$

4. (a) Let $f(x) = 1/x$. Show that f is integrable on $[1, L]$ and $[L^{-1}, 1]$ for any $L > 1$. Define the function $\log : (0, \infty) \rightarrow \mathbb{R}$ by

$$\log(x) = \int_1^x f(t)dt, \quad \text{for any } x > 0.$$

Additional Exercise 4(a.) Let $f(x) = \frac{1}{x}$. Then, define the function $\log : (0, \infty) \rightarrow \mathbb{R}$ by $\log(x) = \int_1^x f(t)dt$, for any $x > 0$. We want to show that f is integrable on $[1, L]$ and $[L^{-1}, 1]$ for any $L > 1$.

Firstly, we want to show that $\frac{1}{x}$ is integrable on $[1, L]$. For this, as $\frac{1}{x}$ is bounded, it suffices to show that for every $\epsilon > 0$ there exists a partition P of $[a, b]$ such that $U(f, P) - L(f, P) < \epsilon$.

Then, by **Definition 13.11**, we have that $M_i(f) = \sup\{f(x) \mid t_{i-1} \leq x \leq t_i\}$ and $m_i(f) = \inf\{f(x) \mid t_{i-1} \leq x \leq t_i\}$. As $\frac{1}{x}$ is decreasing on $[1, L]$, $f(x) = \frac{1}{x}$ we have that $M_i(f) = \frac{1}{t_{i-1}}$ and $m_i(f) = \frac{1}{t_i}$, hence $M_i(f) - m_i(f) = \frac{1}{t_{i-1}} - \frac{1}{t_i} = |f(t_{i-1}) - f(t_i)|$. We note that

$$U(f, P) - L(f, P) = \sum_{i=1}^n (M_i(f) - m_i(f))(t_i - t_{i-1}).$$

Now, consider $f(x) = \frac{1}{x}$. We have that $f(x) : [1, L] \rightarrow \mathbb{R}$ which is continuous and bounded. Then, by **Corollary 13.8**, we have that $f(x)$ is uniformly continuous. Therefore, we have that for all $\epsilon > 0$, there exists a $\delta > 0$ such that for all $t_i, t_{i-1} \in P$ where P is a partition, we have that if $|t_{i-1} - t_i| < \delta$ then $|f(t_{i-1}) - f(t_i)| < \frac{\epsilon}{L-1}$ which is equivalent to $\left| \frac{1}{t_{i-1}} - \frac{1}{t_i} \right| < \frac{\epsilon}{L-1}$.

Now consider partition P . We can define the partition such that every subinterval satisfies $t_i - t_{i-1} < \delta$. On each such subinterval we have $M_i(f) - m_i(f) < \frac{\epsilon}{L-1}$. Therefore

$$U(f, P) - L(f, P) = \sum_{i=1}^n (M_i(f) - m_i(f))(t_i - t_{i-1}) < \frac{\epsilon}{L-1} \cdot (L-1) = \epsilon.$$

Since $\sum_{i=1}^n (t_i - t_{i-1}) = L - 1$. By **Theorem 13.18**, f is integrable on $[1, L]$. This completes the proof.

Now, we want to show that f is integrable on $[L^{-1}, 1]$ for any $L > 1$. In the same light, we have that f is uniformly continuous and bounded on $[L^{-1}, 1]$. Then, we want to show that for all $\epsilon > 0$ there exists a partition P such that $U(f, P) - L(f, P) < \epsilon$. Then, by **Definition 13.11**, we have that, as $\frac{1}{x}$ is decreasing on $[L^{-1}, 1]$, $M_i(f) = \frac{1}{t_{i-1}}$ and $m_i(f) = \frac{1}{t_i}$, hence $M_i(f) - m_i(f) = \frac{1}{t_{i-1}} - \frac{1}{t_i} = |f(t_{i-1}) - f(t_i)|$. We note that

$$U(f, P) - L(f, P) = \sum_{i=1}^n (M_i(f) - m_i(f))(t_i - t_{i-1}).$$

As $f(x)$ is uniformly continuous we have that there exists a $\delta > 0$ such that for all $t_i, t_{i-1} \in P$, if $|t_{i-1} - t_i| < \delta$ then $|f(t_{i-1}) - f(t_i)| < \frac{\epsilon}{1-L^{-1}}$ which is equivalent to $\left| \frac{1}{t_{i-1}} - \frac{1}{t_i} \right| < \frac{\epsilon}{1-L^{-1}}$. We can define the partition such that every subinterval satisfies $|t_i - t_{i-1}| < \delta$. On each such subinterval we have $M_i(f) - m_i(f) < \frac{\epsilon}{1-L^{-1}}$. Therefore

$$U(f, P) - L(f, P) = \sum_{i=1}^n (M_i(f) - m_i(f))(t_i - t_{i-1}) < \frac{\epsilon}{1-L^{-1}} \cdot (1 - L^{-1}) = \epsilon.$$

Since $\sum_{i=1}^n (t_i - t_{i-1}) = 1 - L^{-1}$. By **Theorem 13.18**, f is integrable on $[L^{-1}, 1]$. This completes the proof. $\square$

(b) Prove that, for any $a, b \in (0, \infty)$ we have

$$\log(ab) = \log(a) + \log(b)$$

Hint: First show that

$$\int_1^a \frac{1}{x} dx = \int_b^{ab} \frac{1}{x} dx.$$

Additional Exercise 4(b.) Let $\log(x) = \int_1^x \frac{1}{t} dt$ for any $x > 0$. We want to show that $\log(ab) = \log(a) + \log(b)$.

Firstly, consider the equality $\int_1^a \frac{1}{x} dx = \int_b^{ab} \frac{1}{x} dx$. We want to show this is true. As $f(t) = \frac{1}{t}$ is continuous for all $\epsilon > 0$, we have that there exists $n \in \mathbb{N}$ such that for partition $P = \{b, b + \frac{ab-b}{n}, b + 2\frac{ab-b}{n}, \dots, b + (n-1)\frac{ab-b}{n}, b + ab - b\}$.

Now we want to show that $\int_1^a \frac{1}{x} dx = \int_b^{ab} \frac{1}{x} dx$. For this, we want to compute $L(f, P)$ where $P = \{b, b + \frac{ab-b}{n}, b + 2\frac{ab-b}{n}, \dots, b + (n-1)\frac{ab-b}{n}, ab\}$. By **Definition 13.11**, we have that $L(f, P) = \sum_{k=1}^n \frac{(ab-b)}{n} \cdot f(b + k \cdot \frac{(ab-b)}{n})$. As $f(t) = \frac{1}{t}$, we have that

$$\frac{(ab-b)}{n} \cdot \sum_{k=1}^n \left(\frac{n}{kab + (n-k)b} \right)$$

We can rearrange this equation to get

$$b(a-1) \cdot \frac{1}{b} \sum_{k=1}^n \left(\frac{1}{ka + (n-k)} \right)$$

$$(a-1) \cdot \sum_{k=1}^n \left(\frac{1}{ka + (n-k)} \right)$$

Now, consider $\int_1^a f$. We define the partition $Q = \{1, 1 + \frac{a-1}{n}, 1 + 2 \cdot \frac{a-1}{n}, \dots, 1 + (n-1) \frac{a-1}{n}, a\}$ of $[1, a]$. Then, by **Definition 13.11**, we have that $L(f, Q) = (a-1) \sum_{k=1}^n \frac{1}{ka + n - k}$. We note that $L(f, P) = L(f, Q)$, hence the lower sums of f over $[b, ab]$ and $[1, a]$ coincide for every n .

Therefore, for n large enough we have that $\int_b^{ab} f \geq L(f, P) \geq \int_b^{ab} f - \epsilon$ and $\int_1^a f \geq L(f, Q) \geq \int_1^a f - \epsilon$. Thus, we have that $\int_1^a f = \int_b^{ab} f$.

Now we want to show that $\log(ab) = \log(a) + \log(b)$. We have that, by **Additional Exercise 4(a.)** we can write this equation as

$$\int_1^{ab} f = \int_1^a f + \int_1^b f$$

As we have shown that $\int_1^a f = \int_b^{ab} f$, we can substitute it into the equation to get

$$\int_1^{ab} f = \int_b^{ab} f + \int_1^b f$$

Then, by **Theorem 13.25(a.)** we have that this equation holds. Thus, this suffices to prove that for any $a, b \in (0, \infty)$ we have $\log(ab) = \log(a) + \log(b)$. $\square$

- (c) Show that $\log$ is continuous, unbounded, strictly increasing and the image of $\log$ is $\log((0, \infty)) = \mathbb{R}$.

Additional Exercise 4(c.) We want to show that $\log$ is continuous, unbounded and strictly increasing.

Firstly, we want to show that $\log$ is continuous. We have that $\log(x) = \int_1^x \frac{1}{t} dt$ for any $x > 0$. We have shown that $\frac{1}{x}$ is integrable on $[1, L]$ and $[L^{-1}, 1]$ where $L \in \mathbb{R}$. Therefore, by **Theorem 13.29**, we have that $\int_1^L \frac{1}{t} dt$ is continuous and therefore $\log(x)$ is continuous.

Secondly, we want to show that $\log(x)$ is unbounded. For this, it suffices to show that $\log$ has no upper bound. For this, we can use induction. Suppose for the sake of contradiction that $\log$ has an upper bound. Now, consider the base case where $x = 1$. Then,

we have that $\int_1^1 f = 0$. Then, let M be the upper bound of $\log(x)$. Then, let $x = n$ where in $\int_1^n f(t) dt = M$. Now consider $n + 1$. Then, we have that $\int_1^{n+1} f = N$ where $N > M$. This is a contradiction as M is the upper bound for $\int_1^x f(t) dt$. As this is true for all n , we have that, as $\mathbb{R}$ is unbounded and $x \in \mathbb{R}$, $\log(x)$ must be unbounded.

Thirdly, we want to show that $\log$ is strictly increasing. By **Theorem 14.1**, as $f(t) = \frac{1}{t}$ is continuous, we have that $\log(x) = \int_1^x \frac{1}{t} dt$ is differentiable and $\log'(x) = \frac{1}{x}$. As $x > 0$, we have that $\log'(x) = \frac{1}{x} > 0$ for all $x > 0$. Therefore, by **Corollary 12.16**, $\log$ is strictly increasing.

Fourthly, we want to show that $\log((0, \infty)) = \mathbb{R}$. We have shown that $\log$ is unbounded above and, by a symmetric argument applied to $[L^{-1}, 1]$, unbounded below. Therefore, for any $y \in \mathbb{R}$, there exist $a, b \in (0, \infty)$ such that $\log(a) < y < \log(b)$. Since $\log$ is continuous, by the Intermediate Value Theorem there exists $x \in (a, b)$ such that $\log(x) = y$. Hence $\log((0, \infty)) = \mathbb{R}$. $\square$

5.

Definition 14.8. We say that $\lim_{x \rightarrow \infty} f(x) = L$ if, for every $\epsilon > 0$, there is a real number X such that if $x \geq X$ then $|f(x) - L| < \epsilon$. (There is an analogous definition of limit as $x \rightarrow -\infty$.)

Definition 14.9. Suppose that $f: [a, b] \rightarrow \mathbb{R}$ is integrable for every $b > a$, where a is fixed. If $\lim_{b \rightarrow \infty} \int_a^b f(x) dx$ exists, we denote it by $\int_a^\infty f(x) dx$ and call it an “improper” integral.

a) Prove that $\lim_{x \rightarrow \infty} \frac{1}{x^n} = 0$, for any $n \in \mathbb{N}$.

Additional Exercise 5(a.) Let $f(x) = \frac{1}{x}$. We want to show that $\lim_{x \rightarrow \infty} \frac{1}{x^n} = 0$. For this, it suffices to show that for every $\epsilon > 0$ there is a $X \in \mathbb{R}$ such that if $x \geq X$ then $|f(x) - L| < \epsilon$.

Consider $\frac{1}{x}$. We note that for all $n \in \mathbb{N}$ and $x \geq 1$, we have that $x^n \geq x$, hence $\frac{1}{x^n} \leq \frac{1}{x}$. Now, fix $\epsilon > 0$. Then, as $\epsilon > 0$, we know that there exists some $x \in \mathbb{R}$ such that $\frac{1}{x} < \epsilon$. As $\frac{1}{x}$ is decreasing, we have that there exists some $X \in \mathbb{R}$ such that if $x \geq X$ then $\frac{1}{x} < \epsilon$.

Now consider $|f(x) - L|$. As we want to show that $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$, we want to show $|\frac{1}{x} - 0| < \epsilon$. As $\frac{1}{x} < \epsilon$ for all $x \geq X$ we have that this holds. As this holds for $\frac{1}{x}$, we have that this holds for $\frac{1}{x^n}$ for all $n \in \mathbb{N}$. Lastly, as the epsilon is arbitrary, we have that this statement holds true. $\square$

b) Find $\int_1^\infty \frac{1}{x^2} dx$.

Additional Exercise 5(b.) Let $f(x) = \frac{1}{x^2}$. We want to find $\int_1^\infty \frac{1}{x^2} dx$.

We have that $\frac{1}{x^2}$ is integrable on $[1, L]$ where $L \in \mathbb{R}$. Now suppose there is a function G that is continuous on $[1, L]$ where $L \in \mathbb{R}$ and differentiable on $(1, L)$ such that $f = G'$ on $(1, L)$ where $L \in \mathbb{R}$. By **Theorem 12.10**, we have that $G' = \frac{1}{x^2}$ and hence $G = -\frac{1}{x}$.

By **Theorem 14.5**, we have that $\int_1^\infty \frac{1}{x^2} dx$ can be written as

$$\int_1^\infty \frac{1}{x^2} dx = \lim_{L \rightarrow \infty} \int_1^L \frac{1}{x^2} dx = \lim_{L \rightarrow \infty} (G(L) - G(1)) = \lim_{L \rightarrow \infty} \left(-\frac{1}{L} + 1 \right) = 1$$

where the last step follows by **Additional Exercise 5(a)**.

Hence, this works to show that $\int_1^\infty \frac{1}{x^2} dx = 1$. □

Additional Exercise 5(c.) Let $f(x) = \frac{1}{x}$. We want to show that $\int_1^\infty \frac{1}{x} dx$ does not exist.

By **Additional Exercise 4** we have that $\int_1^a \frac{1}{x} dx = \log(a)$ for all $a > 1$. Then, fix the lower bound as 1. As $a \rightarrow \infty$, $\log(a) \rightarrow \infty$, so $\lim_{a \rightarrow \infty} \int_1^a \frac{1}{x} dx$ does not exist as a real number. Therefore, we have that $\int_1^\infty \frac{1}{x} dx$ is an improper integral and does not exist. This completes the proof. □

6. In this problem, you may assume that $\sin$, $\cos$, and $\arctan$ are differentiable functions with domain $\mathbb{R}$ satisfying $\frac{d}{dx} \sin x = \cos x$, and $\frac{d}{dx} \cos x = -\sin x$, and $\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$. You may also assume that $\frac{d}{dx}(x^\alpha) = \alpha x^{\alpha-1}$, for any $\alpha \in \mathbb{R}$.

(a) Find $\int_1^2 t^2 \sqrt{t^3 - 1} dt$.

(b) Find $\frac{d}{dx} \int_0^1 \frac{(\sin t)(\cos t)}{t^3 + 1} dt$.

(c) Find $\frac{d}{dx} \int_0^{\cos x} \frac{1}{t^4 + 1} dt$.

(d) For $a, b \in \mathbb{R}$, find $\int_a^b \frac{\cos x}{\sqrt[4]{\sin x + 9}} dx$.

(e) Find $\int_0^{\pi/2} \left(\frac{\pi}{2} - x \right) (\cos x) dx$.

(f) Find $\int_0^x t(\arctan t) dt$.

7. In this exercise you may assume familiarity with the trigonometric functions, the logarithmic/exponential functions and their derivatives. Compute the following integrals:

(a) $\int \frac{x dx}{3 - 2x^2},$

(b) $\int \frac{x dx}{4 + x^4},$

(c) $\int \frac{e^x dx}{2 + e^x},$

(d) $\int \tan(x) dx,$

(e) $\int \frac{dx}{1 + e^x},$

(f) $\int_{-1}^1 \frac{x dx}{x^2 + x + 1},$

(g) $\int_0^a x^2 \sqrt{a^2 - x^2} dx.$